# An interpretable tensor model for identifying atrial fibrillation in sinus-rhythm ECG: block-term tensor regression on PTB-XL, benchmarked against deep and foundation models

Dries Cornelissen[1] Axel Faes[2,*]

[1] Data Science Institute (DSI), Hasselt University, Hasselt, Belgium.
[2] Biomedical Data Sciences, Data Science Institute (DSI) and Biomedical Research Institute (BIOMED), Hasselt University, Hasselt, Belgium.

[*] Corresponding author: Axel Faes, axel.faes@uhasselt.be

**Manuscript facts.** Word count from the title page through the figure legends, references excluded: 5806. Abstract word count: 199. Number of figures: 2. Number of tables: 3.

**Author contributions.** Dries Cornelissen: Data curation, Investigation. Axel Faes: Conceptualization, Methodology, Software, Formal analysis, Validation, Visualization, Supervision, Writing (original draft), Writing (review and editing). Both authors approved the final manuscript.

**Abstract**

Atrial fibrillation may leave a substrate that an electrocardiogram in sinus rhythm already carries. Models that read it are usually deep networks trained on private cohorts that do not expose what they use. We test whether an intrinsically interpretable tensor model can identify atrial fibrillation in sinus rhythm on public data. On PTB-XL, sinus-rhythm records followed by an atrial-fibrillation record within 31 days were compared with age- and sex-matched controls over ten control draws. We tensorize the R-peak-aligned median beat into a records-by-time-by-lead array and fit block-term tensor regression (BTTR), whose parameters are readable time-by-lead weight maps. BTTR reached a mean AUROC of 0.597 on the AFIB cohort and 0.643 on the AFIB-or-flutter cohort, and no baseline had a higher mean AUROC. Over the ten draws its AUROC was higher than that of a convolutional network and a fixed-setting HuBERT-ECG foundation-model probe, with pooled paired 95% confidence intervals excluding zero in both cohorts. Its mean advantage over a tuned HuBERT-ECG probe and clinical features had intervals including zero, and it was comparable to an unsupervised CP surrogate. BTTR required beat alignment. The per-lead weight profile reproduced across control draws (Pearson $r = 0.857$), with weight distributed across the beat.

**Keywords:** Atrial Fibrillation; Electrocardiography; Machine Learning; Interpretable machine learning; Block-term tensor regression

# 1 Introduction

Atrial fibrillation is the most common sustained arrhythmia and a leading cause of ischemic stroke. Much of its burden falls on patients who are in sinus rhythm at the moment of recording, because the arrhythmia is paroxysmal and intermittent. A standard screening electrocardiogram captured between episodes can therefore look entirely normal while the underlying atrial substrate is already present. Reading that substrate from a single sinus-rhythm trace would let clinicians flag patients for monitoring before an event. A convolutional network trained on a large institutional cohort first showed that atrial fibrillation can be identified from a sinus-rhythm electrocardiogram [2].

That result has been replicated and extended across cohorts, populations and device classes [3, 8, 11, 18, 21]. The models are accurate, and the clinical signal is real. Two features limit what the field can learn from them. First, they rest on deep supervised learning on private data, which makes head-to-head comparison between methods difficult and external scrutiny harder still. Second, they are end-to-end black-box predictors whose performance is reported and whose internal representation is not: a clinician is told that a normal-looking trace carries risk but not which part of the beat, or which leads, the model reacts to. For a finding that contradicts the visible reading of the electrocardiogram, the absence of an account is a real obstacle to trust and to physiological interpretation.

The usual response is a post-hoc attribution (saliency, gradient, attention or surrogate) fitted to a trained network after the fact [12–14, 17]. These estimate what a fixed black box might have used and leave what it used unchanged, different attribution methods can disagree on the same model, and their stability and fidelity are active concerns [9]. Intrinsic interpretability takes the alternative position that the model's parameters should be the explanation.

Tensor models take that position. Arranging the multilead beat as a records-by-time-by-lead array turns the electrocardiogram into a structured signal on which a supervised low-rank model can be fit directly, and its fitted factors, the per-block time-by-lead weight maps, are the decision itself and read as a spatio-temporal signature rather than an approximation of one. Tensor decompositions have a long history on the electrocardiogram, almost entirely in unsupervised source

separation: extracting atrial activity, separating maternal and fetal signals, and related blind decompositions [5, 6, 16]. Supervised block-term modeling, where the decomposition is fit to a label, has been developed on the electroencephalogram [19], and block-term tensor regression (BTTR) was introduced in that context for a different problem [7]. We bring this estimator to the identification of atrial fibrillation in sinus rhythm and benchmark it openly on public data.

We study this task on PTB-XL [20], a public clinical electrocardiogram database, against age- and sex-matched controls so that age does not carry the label. Beat alignment before tensorization has been examined for ECG learning [15], and an alignment ablation measures what R-peak alignment of the median beat contributes to BTTR and to a convolutional network (CNN). We evaluate BTTR against clinical features, a CNN in the style of [2], a capacity-matched unsupervised CP surrogate, and frozen embeddings of HuBERT-ECG [4], a foundation model pretrained on large unlabeled electrocardiogram corpora and used as an existing estimator. All methods share the cohorts, patient-grouped folds and ten matched control draws, and comparisons use a paired patient-level bootstrap.

The contributions are as follows.

1. We present, to our knowledge, the first intrinsically interpretable tensor model for identifying atrial fibrillation in sinus rhythm, evaluated on public data with the record-level cohort and split manifests released with the code.

2. Under matched controls, no baseline has a higher ten-draw mean AUROC than BTTR (0.597 on the AFIB cohort, 0.643 on the AFIB-or-flutter cohort). Its AUROC is higher than that of a CNN and of a fixed-setting frozen HuBERT-ECG probe with pooled paired 95% confidence intervals excluding zero in both cohorts, the comparisons with a tuned HuBERT-ECG probe and clinical features do not separate the methods, and BTTR is comparable to an unsupervised CP surrogate fit to the same tensor.

3. BTTR exposes its decision as eight time-by-lead weight maps, with a per-lead weight profile that reproduces across control draws (Pearson $r = 0.857$) and weight distributed across the beat, the largest share in the T-wave window.

4. BTTR requires R-peak registration: its aligned-minus-unaligned AUROC difference has a pooled paired 95% CI excluding zero in both cohorts, and without registration its AUROC falls to chance.

# 2 Methods

## 2.1 Data and cohorts

We use PTB-XL 1.0.3 [20], a public clinical 12-lead electrocardiogram database of 21,799 records with patient identifiers, SCP rhythm codes and recording dates. We use the 100 Hz records. A record is positive if it is a sinus-rhythm record (SCP code SR) of a patient who has an atrial-fibrillation record (AFIB) dated 0 to 31 days after it, so the atrial-fibrillation record always follows the sinus-rhythm record. Every such sinus-rhythm record is a positive, and a patient can contribute more than one. The primary cohort, AFIB, has 42 positive records from 37 patients. The secondary cohort, AFIB or flutter, counts an AFIB or AFLT record as the event and has 55 positive records from 46 patients. The median interval from the sinus-rhythm record to the first atrial-fibrillation record in the window is 5 days, and 7.1% (AFIB) and 14.5% (AFIB or flutter) of positives share the day. The task is therefore concurrent identification. The arrhythmia is recorded within days of

the sinus-rhythm trace, which is not long-horizon prediction of future onset, and PTB-XL carries no longitudinal follow-up.

Control candidates are patients with no AFIB and no AFLT record anywhere in PTB-XL. Each contributes one record, its first sinus-rhythm record (lowest ecg_id), which gives 14,899 candidates. Controls are matched 1:3 without replacement. Positives are processed in a seeded random order, and each is assigned the three unused candidates of the same sex with the nearest age, with ties broken by a seeded random jitter. PTB-XL codes ages above 89 as 300, so such records match within that code. This gives 126 controls for the AFIB cohort and 165 for the AFIB-or-flutter cohort, a prevalence of 0.25. On the seed-0 draw, positives and controls are identical in median age (75 years; IQR 64 to 81 for AFIB and 64 to 84 for AFIB or flutter), female fraction (0.429 and 0.382) and fraction over 89 years (0.024 and 0.018). Age alone used as a score gives AUROC 0.500 on every draw. Ten control draws (seeds 0 to 9) rerun the matching with that seed while the positives stay fixed. All folds are patient-grouped, so all records of a patient fall in one fold. For every draw of both cohorts, each record's ecg_id, patient identifier, label, age, sex, recording date, days to the atrial-fibrillation record and outer fold are deposited with the code at `https://github.com/TheAxeC/bttr-af-ecg`, and one command regenerates all results from the PTB-XL files.

## 2.2 Preprocessing and tensorization

Each lead is filtered with a zero-phase second-order Butterworth high-pass at 0.5 Hz and a 50 Hz notch ($Q = 30$), denoised by a sym4 wavelet decomposition to level 3 with soft thresholding of the detail coefficients at 7 times the standard deviation of the finest detail level, and then centred and scaled by its maximum absolute value. These are fixed per-record operations with no learned parameters. R-peaks are detected on lead II by an energy-envelope detector with a 150 ms integration window and a 250 ms refractory distance. The median beat of each lead is the median over all complete beats of the window from $-250$ to $+400$ ms around each R-peak, which at 100 Hz gives 65 samples. Stacking the 12 leads gives a records-by-65-by-12 tensor, the input to BTTR, the CP surrogate and the aligned CNN. The unaligned condition of the ablation uses the cleaned 10 s strip without R-peak registration.

## 2.3 Models

**BTTR.** Block-term tensor regression [7] is a supervised, inductive, deflation-based estimator. The binary label is mapped to a $\pm 1$ target and fit by least squares, giving a linear decision score rather than a logistic one. The model proceeds in $K$ iterations. At each iteration a block is extracted, and both the data-tensor residual and the target residual are deflated before the next block is fit, so each block accounts for structure the previous blocks left behind. Each block is a rank-$(L_2, L_3, 1)$ term of the cross-covariance tensor between the data and the label: the rank in the label mode is fixed at 1, the tensor-vector case for a vector label, and automatic component estimation selects the time-mode rank $L_2$ and the lead-mode rank $L_3$ of each block within the training fold over signal-to-noise thresholds from 20 to 40 dB in steps of 5 dB and energy ratios of 95, 97 and 99%. Each block's time-by-lead weight map therefore has rank up to $\min(L_2, L_3)$. The model is a supervised, label-fit sum of block terms, and each block's time-by-lead weight map is directly readable. The block count was fixed at $K = 8$ before the benchmark. A nested check that tunes $K$ inside the training folds is reported in Results.

**Clinical features.** Seven features from lead II feed a logistic regression after a training-fold standard scaler: the mean RR interval, the standard deviation of RR, RMSSD, the RR coefficient of variation, pNN50, a PR-segment-to-QRS energy ratio as a P-wave proxy, and the median peak-to-peak amplitude. This baseline encodes the standard electrophysiological reading.

**CP surrogate.** An unsupervised rank-8 PARAFAC decomposition (tensorly, $\ell_2$ regularisation $10^{-3}$, at most 100 iterations, deterministic truncated-SVD initialisation) is fit to the training fold only. Test records are projected onto the fixed training time and lead factors by least squares, and the standardised coordinates feed a logistic regression. The rank matches the BTTR block count so that capacity is held constant, and the contrast with BTTR isolates the effect of fitting the decomposition to the label.

**CNN.** A one-dimensional CNN in the style of [2] has two convolution blocks (32 filters of width 7 with batch normalisation, ReLU, max-pooling and dropout 0.3; 64 filters of width 5 with batch normalisation, ReLU and global average pooling), dropout 0.5 and a linear output. It is trained with class-weighted binary cross-entropy and Adam (learning rate $10^{-3}$, weight decay $10^{-3}$) for up to 120 epochs, with early stopping (patience 18) on a patient-grouped validation split carved from the training fold, and three seeds are ensembled. Its input is the median-beat tensor in the aligned condition and the strip in the unaligned condition. It stands in for the deep-learning approach at a scale appropriate to the cohort size.

**HuBERT-ECG probe.** HuBERT-ECG [4] is a self-supervised electrocardiogram foundation model pretrained on 9.1 million recordings. We use the checkpoint `Edoardo-BS/hubert-ecg-small`, held frozen without fine-tuning. Its input is the first 5 s of the 12 cleaned leads at 100 Hz, concatenated and z-normalised per record, and its mean-pooled 512-dimensional embedding feeds a linear probe made of a training-fold standard scaler and an L2-regularised logistic regression. The tuned probe selects $C$ from $\{0.001, 0.01, 0.1, 1\}$ by patient-grouped 3-fold inner cross-validation within each training fold, and the fixed-setting probe uses $C = 0.1$. Neither probe sets $C$ on test data. The self-supervised pretraining corpus of HuBERT-ECG includes PTB-XL [1]. The pretraining used no labels, so the frozen encoder may have seen the unlabelled benchmark records, and any such exposure would favour the probe.

## 2.4 Evaluation

Two evaluation protocols are used. Under protocol P1, each of the ten control draws is evaluated by patient-grouped stratified 5-fold cross-validation with shuffling and the fold seed equal to the draw seed. The AUROC is computed on the pooled out-of-fold scores of that draw, and the ten AUROCs are summarised as the mean ± population standard deviation (SD) across the draws, with the range. Every method, BTTR and the CNN included, is evaluated under P1 on the same draws and folds. Under protocol P2, the seed-0 draw and its fold split are evaluated once. The AUROC is the point estimate on the pooled out-of-fold scores, and its 95% confidence interval (CI) is the 2.5th to 97.5th percentile range of 2000 patient-level bootstrap resamples, each drawing patients with all their records. The seed-0 draw was fixed before analysis as the code default. On the AFIB cohort it is the draw on which BTTR scores highest of the ten, so P1 is the primary summary, and P2 supplies the marginal CIs, the AUPRC and a secondary paired comparison on one pre-fixed draw.

Method comparisons use paired patient-level bootstraps, and the primary comparison is on P1. In each of 2000 iterations, one patient resample is drawn with replacement from the union

of the patients of the ten draws (1073 for the AFIB cohort, 1309 for the AFIB-or-flutter cohort) and applied to every draw, with the records of each drawn patient kept with their multiplicity. The AUROC difference BTTR minus baseline is computed per draw on the out-of-fold scores and averaged over the ten draws. The same procedure gives the alignment effect, aligned minus unaligned AUROC, for BTTR and for the CNN. The point estimate is the P1 mean difference on the full scores, the 95% CI is the 2.5th to 97.5th percentile range of the 2000 resampled means, and the fraction is the share of resamples with a positive mean difference. Resampling patients jointly across draws keeps the dependence that arises because the draws share their positives. The secondary comparison is on P2: in each of 2000 resamples of the seed-0 draw, the difference is computed on the same resampled patients, and the reported difference is the point estimate on the full out-of-fold scores with its percentile CI. A positive point estimate or a fraction above 0.5 does not establish superiority, equivalence or non-inferiority, and no equivalence or non-inferiority margin was prespecified. Where the CI excludes zero, we report that BTTR's AUROC was higher. Where it includes zero, we report that the comparison does not separate the two methods. We also report the number of draws, of ten, on which BTTR's AUROC exceeds the baseline's. A paired difference is the appropriate object here, because the overlap of two marginal intervals ignores the correlation between methods evaluated on the same patients.

Every learned step is fit on the training rows of each outer fold only: the standard scalers, the CP factors, the logistic regressions, the BTTR component estimation and fit, the CNN training and early stopping, and the tuned $C$ of the HuBERT-ECG probe. Signal preprocessing is per record with fixed settings. $K = 8$ and the CP rank were fixed before the benchmark.

Finally, to calibrate task difficulty, we include a detection-context cohort of overt sinus rhythm versus overt atrial fibrillation, 150 records per class, evaluated by one patient-grouped 5-fold cross-validation. This analysis uses the 500 Hz median beat with three RR pseudo-leads appended. It is illustrative and supports no accuracy claim of its own.

# 3 Results

## 3.1 Identifying atrial fibrillation in sinus rhythm

Table 1 reports AUROC under P1 for all methods and for age alone, with the P2 values of the seed-0 draw in its lower panel. Fig. 1 shows the P2 point estimates with their 95% CIs, and Table 2 gives the paired comparisons.

Under P1, BTTR reaches a mean AUROC of $0.597 \pm 0.035$ on the AFIB cohort and $0.643 \pm 0.026$ on the AFIB-or-flutter cohort. The CP surrogate reaches $0.592 \pm 0.032$ and $0.634 \pm 0.031$, the tuned HuBERT-ECG probe $0.548 \pm 0.040$ and $0.606 \pm 0.028$, clinical features $0.537 \pm 0.052$ and $0.562 \pm 0.018$, the CNN $0.495 \pm 0.057$ and $0.522 \pm 0.038$, and the HuBERT-ECG probe with $C = 0.1$ $0.488 \pm 0.039$ and $0.537 \pm 0.049$. Age alone gives 0.500 on every draw, so the matching removes the age signal. Under P2, BTTR reaches 0.673 [0.576, 0.770] on the AFIB cohort and 0.621 [0.532, 0.708] on the AFIB-or-flutter cohort, with AUPRC 0.484 and 0.336 against a prevalence of 0.25.

Over the ten draws (P1), BTTR's AUROC was higher than that of the CNN (+0.102 [+0.027, +0.177] on the AFIB cohort and +0.121 [+0.044, +0.194] on the AFIB-or-flutter cohort) and of the HuBERT-ECG probe with $C = 0.1$ (+0.109 [+0.031, +0.187] and +0.106 [+0.027, +0.187]), with each pooled paired CI excluding zero. BTTR was higher than the CNN on 9 and 10 of the ten draws and higher than the $C = 0.1$ probe on all ten in both cohorts. Against the tuned HuBERT-ECG probe (+0.049 [−0.027, +0.123] and +0.037 [−0.036, +0.115]) and clinical features (+0.060 [−0.034, +0.159] and +0.082 [−0.005, +0.172]), the mean difference was positive and the pooled CI included zero, so these comparisons do not separate the methods. BTTR was higher than the

**Table 1** Identification of atrial fibrillation in sinus rhythm under age- and sex-matched controls. Upper panel, protocol P1: AUROC as mean ± SD across the ten control draws, with the range in brackets. Lower panel, protocol P2 (seed-0 draw): AUROC point estimate with its 95% patient-level bootstrap CI, and AUPRC in parentheses. The prevalence, which is the AUPRC of a random score, is 0.25 in both cohorts. The seed-0 draw was fixed in advance and is the highest-scoring of the ten draws for BTTR on the AFIB cohort. HuBERT-ECG probe: frozen foundation-model embedding with a logistic probe, its $C$ tuned in inner folds or fixed at 0.1. The tuned probe has no marginal P2 interval; its P2 comparison is the paired row in Table 2. CNN: Attia-style one-dimensional network. CP surrogate: unsupervised decomposition of the same tensor as BTTR

| Method | AFIB (42 pos, 126 ctrl) | AFIB or flutter (55 pos, 165 ctrl) |
|---|---|---|
| *P1: AUROC mean ± SD [range]* | | |
| BTTR | $0.597 \pm 0.035$ [0.555, 0.673] | $0.643 \pm 0.026$ [0.605, 0.699] |
| CP surrogate | $0.592 \pm 0.032$ [0.552, 0.663] | $0.634 \pm 0.031$ [0.589, 0.679] |
| HuBERT-ECG probe, tuned $C$ | $0.548 \pm 0.040$ [0.472, 0.605] | $0.606 \pm 0.028$ [0.564, 0.654] |
| clinical features | $0.537 \pm 0.052$ [0.450, 0.638] | $0.562 \pm 0.018$ [0.537, 0.591] |
| CNN | $0.495 \pm 0.057$ [0.407, 0.586] | $0.522 \pm 0.038$ [0.438, 0.569] |
| HuBERT-ECG probe, $C = 0.1$ | $0.488 \pm 0.039$ [0.423, 0.543] | $0.537 \pm 0.049$ [0.470, 0.615] |
| age alone | 0.500 on every draw | 0.500 on every draw |
| *P2: AUROC [95% CI] (AUPRC)* | | |
| BTTR | 0.673 [0.576, 0.770] (0.484) | 0.621 [0.532, 0.708] (0.336) |
| CP surrogate | 0.663 [0.562, 0.764] (0.463) | 0.610 [0.523, 0.694] (0.312) |
| clinical features | 0.513 [0.419, 0.614] (0.264) | 0.591 [0.510, 0.673] (0.321) |
| CNN | 0.444 [0.343, 0.552] (0.230) | 0.510 [0.413, 0.602] (0.262) |
| HuBERT-ECG probe, tuned $C$ | see Table 2 | see Table 2 |
| HuBERT-ECG probe, $C = 0.1$ | 0.478 [0.373, 0.588] (0.284) | 0.594 [0.506, 0.678] (0.384) |

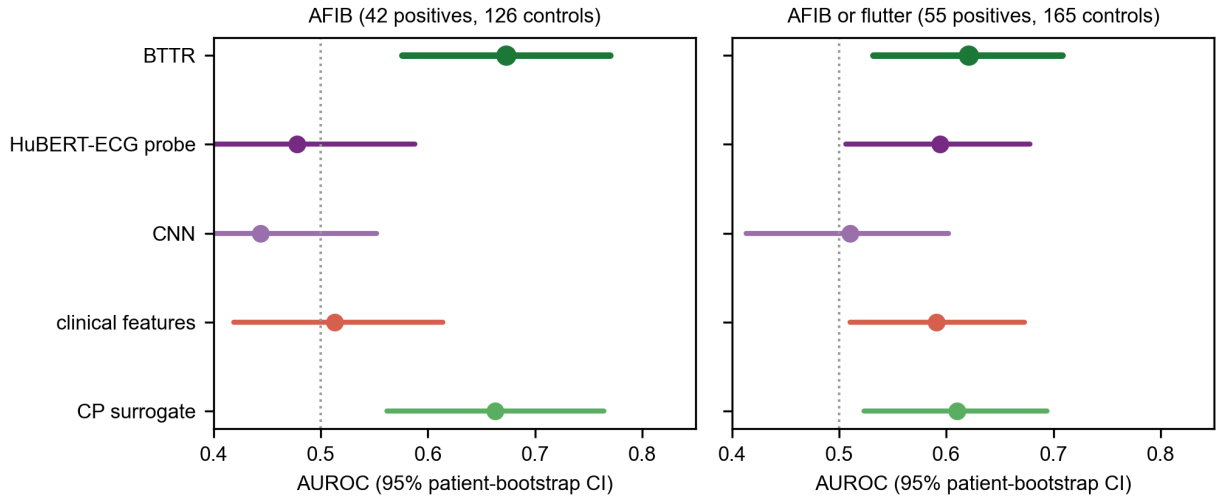



**Fig. 1** AUROC under protocol P2 (seed-0 draw) with 95% patient-level bootstrap confidence intervals for all methods on both cohorts. The dotted line marks chance, and the HuBERT-ECG probe shown uses $C = 0.1$. The ten-draw summary (P1) is in Table 1 and the paired comparisons are in Table 2

**Table 2** Paired comparison of BTTR with each baseline, as the AUROC difference BTTR minus baseline. Primary columns, P1 over the ten control draws: the mean difference with its pooled paired 95% CI, from 2000 patient resamples drawn from the union of the patients of all draws and applied to every draw; and the fraction of resamples with a positive mean difference. The number of draws on which BTTR is higher is given in the text. Secondary column, P2 on one pre-fixed draw (seed 0): the difference on the out-of-fold scores with its 95% paired patient-level bootstrap CI. A positive difference or a fraction above 0.5 does not establish superiority, equivalence or non-inferiority, and no margin was prespecified

| | P1, ten draws (primary) | | P2, seed-0 draw |
|---|---|---|---|
| Baseline | Mean $\Delta$ [pooled 95% CI] | Fraction | $\Delta$ [95% CI] |
| *AFIB cohort* | | | |
| HuBERT-ECG probe, tuned $C$ | +0.049 [−0.027, +0.123] | 0.91 | +0.166 [+0.065, +0.268] |
| HuBERT-ECG probe, $C = 0.1$ | +0.109 [+0.031, +0.187] | 1.00 | +0.195 [+0.100, +0.295] |
| CNN | +0.102 [+0.027, +0.177] | 1.00 | +0.229 [+0.128, +0.324] |
| clinical features | +0.060 [−0.034, +0.159] | 0.88 | +0.159 [+0.025, +0.296] |
| CP surrogate | +0.004 [−0.038, +0.047] | 0.57 | +0.010 [−0.039, +0.056] |
| *AFIB-or-flutter cohort* | | | |
| HuBERT-ECG probe, tuned $C$ | +0.037 [−0.036, +0.115] | 0.83 | +0.029 [−0.071, +0.130] |
| HuBERT-ECG probe, $C = 0.1$ | +0.106 [+0.027, +0.187] | 0.99 | +0.027 [−0.073, +0.130] |
| CNN | +0.121 [+0.044, +0.194] | 1.00 | +0.111 [−0.003, +0.230] |
| clinical features | +0.082 [−0.005, +0.172] | 0.97 | +0.030 [−0.082, +0.145] |
| CP surrogate | +0.009 [−0.033, +0.052] | 0.66 | +0.011 [−0.054, +0.074] |

tuned probe on 7 and 9 of the ten draws, and higher than clinical features on 7 and 10. Against the CP surrogate the mean difference was +0.004 [−0.038, +0.047] and +0.009 [−0.033, +0.052], and BTTR was higher on 6 of the ten draws in each cohort. BTTR and the CP surrogate, which use the same tensor, therefore have comparable discrimination. The property BTTR adds is that its decomposition is fit to the label, so its block maps are the weights of the decision itself. On the pre-fixed seed-0 draw (P2), the paired CIs exclude zero against both HuBERT-ECG probes, the CNN and clinical features on the AFIB cohort, and they include zero against every baseline on the AFIB-or-flutter cohort.

The block count was fixed at $K = 8$ before the benchmark. Under P1, sweeping $K$ on the AFIB cohort gives mean AUROC 0.502, 0.536, 0.597, 0.621, 0.596 and 0.573 for $K = 4, 6, 8, 12, 16, 24$, and on the AFIB-or-flutter cohort 0.542, 0.574, 0.643, 0.648, 0.645 and 0.641. $K = 12$ gives a slightly higher mean on both cohorts, $K = 16$ and $K = 24$ give similar values on the AFIB-or-flutter cohort and lower values on the AFIB cohort, and $K = 8$ keeps the set of maps small. As a robustness check, a nested procedure tunes $K$ over $\{6, 8, 12, 16\}$ in inner folds of each training fold. Over the ten draws it gives $0.602 \pm 0.050$ on the AFIB cohort and $0.642 \pm 0.038$ on the AFIB-or-flutter cohort, against $0.537 \pm 0.052$ and $0.562 \pm 0.018$ for clinical features. Appending three RR pseudo-leads to the median beat gives $0.471 \pm 0.036$ and $0.548 \pm 0.032$, below the plain median beat.

### 3.2 Alignment is required for BTTR

Table 3

Table 3 reports the alignment ablation for BTTR and the CNN, each under both protocols on the same draws and folds as the main benchmark. For BTTR, alignment makes the difference between a working model and chance. Under P1 the aligned median beat gives $0.597 \pm 0.035$ on the AFIB cohort and $0.643 \pm 0.026$ on the AFIB-or-flutter cohort, and the unaligned strip gives $0.490 \pm 0.071$ and $0.457 \pm 0.045$. BTTR's aligned AUROC is higher than its unaligned AUROC, with a pooled paired difference of +0.107 [+0.019, +0.195] on the AFIB cohort and +0.186 [+0.097, +0.271] on the AFIB-or-flutter cohort, each CI excluding zero, and the aligned input is higher on all ten draws in both cohorts. The P2 values follow the same pattern: 0.673 against 0.544 and 0.621 against 0.475. The low-rank tensor structure depends on beats being registered.

The CNN stays near chance with or without alignment under matched controls. Under P1 it reaches $0.495 \pm 0.057$ aligned and $0.516 \pm 0.032$ unaligned on the AFIB cohort, and $0.522 \pm 0.038$ and $0.530 \pm 0.032$ on the AFIB-or-flutter cohort. Its pooled paired difference, aligned minus unaligned, is −0.021 [−0.062, +0.019] and −0.008 [−0.053, +0.039], with each CI including zero, and the aligned input is higher on 4 of the ten draws in both cohorts. The ablation therefore establishes that BTTR requires registration. Because the CNN does not discriminate in either condition, it carries no conclusion about alignment for the CNN.

### 3.3 Interpretability: stable and faithful substrate maps

The block maps are model parameters, so they are faithful to the decision by construction; there

Fig. 2

is no separate attribution step whose fidelity could be questioned. Fig. 2 shows the eight maps of BTTR with $K = 8$ fit on the whole seed-0 AFIB cohort.

Stability is assessed across control draws. For each of the ten draws, BTTR with $K = 8$ is fit on the whole cohort of that draw. Its per-lead profile is the sum over blocks of the absolute map weighted by the absolute block coefficient, averaged over time and normalised to sum to one. Taking absolute values removes the sign indeterminacy of each block, since a map and its coefficient can flip sign together. Summing over blocks removes the arbitrary block order, and the normalisation removes scale. The stability is the mean Pearson correlation of the profiles over the 45 pairs of

**Table 3** Alignment ablation for BTTR and the CNN. Both models are evaluated under P1 (AUROC mean ± SD across the ten control draws) and under P2 (seed-0 draw, AUROC point estimate with 95% patient-level bootstrap CI), on the same draws and folds. Aligned: the R-peak-aligned median beat. Unaligned: the cleaned 10 s strip without registration. Lower panel: aligned minus unaligned AUROC under P1, as the mean difference over the ten draws with its pooled paired 95% CI, computed as for Table 2, and the fraction of resamples with a positive mean difference in parentheses

| | | AFIB | | AFIB or flutter | |
|---|---|---|---|---|---|
| Model | Input | P1 | P2 [95% CI] | P1 | P2 [95% CI] |
| BTTR | aligned | $0.597 \pm 0.035$ | 0.673 [0.576, 0.770] | $0.643 \pm 0.026$ | 0.621 [0.532, 0.708] |
| BTTR | unaligned | $0.490 \pm 0.071$ | 0.544 [0.454, 0.635] | $0.457 \pm 0.045$ | 0.475 [0.386, 0.566] |
| CNN | aligned | $0.495 \pm 0.057$ | 0.444 [0.343, 0.552] | $0.522 \pm 0.038$ | 0.510 [0.413, 0.602] |
| CNN | unaligned | $0.516 \pm 0.032$ | 0.528 [0.412, 0.644] | $0.530 \pm 0.032$ | 0.566 [0.468, 0.657] |
| *P1, aligned minus unaligned: mean difference [pooled paired 95% CI] (fraction)* | | | | | |
| BTTR | | +0.107 [+0.019, +0.195] (0.99) | | +0.186 [+0.097, +0.271] (1.00) | |
| CNN | | −0.021 [−0.062, +0.019] (0.16) | | −0.008 [−0.053, +0.039] (0.37) | |

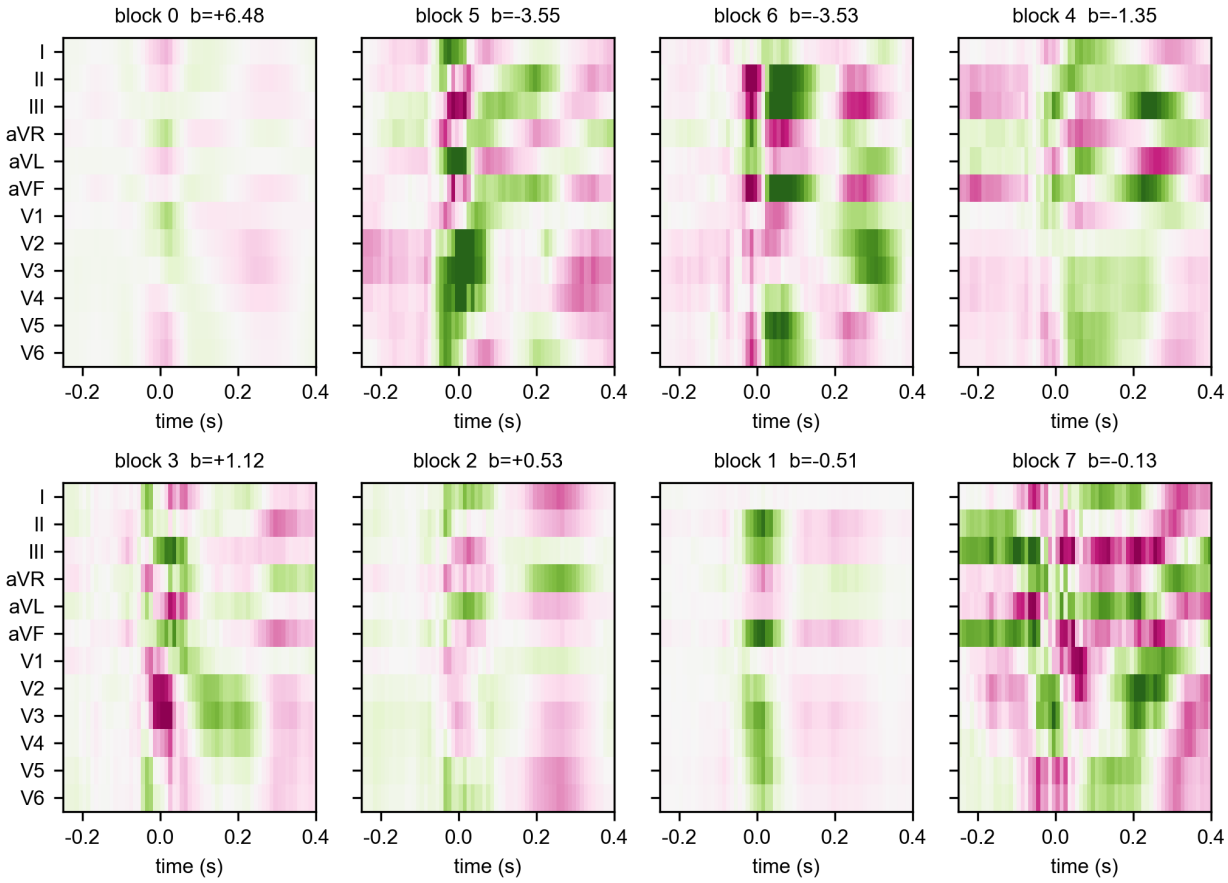



**Fig. 2** BTTR block weight maps of the seed-0 AFIB cohort with $K = 8$, fit on the whole cohort. Each panel is one block's 65-sample by 12-lead weight map, titled with its scalar coefficient $b$ and ordered by $|b|$. Color encodes the signed weight. The maps are fitted model parameters. Across the ten control draws the per-lead weight profile reproduces at Pearson $r = 0.857$

draws. On the AFIB cohort it is $r = 0.857 \pm 0.083$, and the consensus profile puts the largest weight on aVF (0.11), III (0.11), II (0.10) and V3 (0.09). The per-lead profile is thus reproducible across control draws.

Reading where in the beat the weight concentrates shows that it is distributed across the cardiac cycle. We split the window into an atrial window from −250 to −50 ms (P-wave and PR segment), a QRS window from −50 to +50 ms, an ST window from +50 to +150 ms and a T window from +150 to +400 ms, and report each window's share of the summed absolute map weighted by the absolute block coefficient. For BTTR with $K = 8$ fit on the whole seed-0 draw of each cohort, the fit whose AFIB maps Fig. 2 shows, the atrial window holds 0.161 of the weight on the AFIB cohort, the QRS window 0.275, the ST window 0.190 and the T window 0.374. The lead profile spans the inferior leads and the mid-precordium. Established sinus-rhythm atrial markers, such as P-wave indices [10], target the atrial component, which here accounts for a minority of the weight. The model reports that its decision draws on morphology across the whole beat, with the largest share in the T window.

Including flutter in the event definition leaves the atrial share similar: 0.176 on the AFIB-or-flutter cohort against 0.161 on the AFIB cohort. We confine the interpretability claims to the faithfulness of the maps, the reproducibility of the per-lead profile across control draws and the measured weight distribution, and we do not assert an electrophysiological mechanism behind them.

### 3.4 Detection context

On the overt detection cohort, sinus rhythm against atrial fibrillation at 150 records per class under one patient-grouped 5-fold cross-validation on the 500 Hz median beat with RR pseudo-leads, BTTR reaches 0.961 [0.936, 0.982], clinical features 0.972 [0.948, 0.991], and the CP surrogate 0.967 [0.941, 0.987]. Deep-learning detection of overt atrial fibrillation runs near ceiling [17, 20], and the methods cluster together here. The contrast places the identification task, where the rhythm at recording is sinus, well below the overt task in difficulty.

## 4 Discussion

The central result is a supervised tensor model that discriminates sinus-rhythm records followed by atrial fibrillation from age- and sex-matched controls and exposes its decision as readable time-by-lead weight maps. Over the ten control draws no baseline had a higher mean AUROC. BTTR's AUROC was higher than that of the CNN and the fixed-setting HuBERT-ECG probe, with pooled paired 95% CIs excluding zero in both cohorts. Its mean advantage over the tuned HuBERT-ECG probe and clinical features (+0.037 to +0.082) had CIs including zero, so these comparisons do not separate the methods, and its discrimination was comparable to that of the CP surrogate fit to the same tensor. The property BTTR adds over the CP surrogate is that its maps are fit to the label. Any exposure of the HuBERT-ECG encoder to the unlabelled PTB-XL records during pretraining would favour the probes, so it does not account for their lower mean AUROC.

The map analysis adds a finding that the readable model made visible. The weight is distributed across the beat: the T window holds the largest share (0.374) and the atrial window 0.161. We present this as a data-driven observation. It is plausibly consistent with ventricular involvement in the atrial-arrhythmia substrate, but we state that only as plausible; the model reports where its weight sits, not why the physiology puts it there.

The alignment result has a practical consequence for tensor models of the electrocardiogram. BTTR discriminates only on registered beats, so R-peak alignment is a required part of the method.

**Scope.** Four limits bound the claims. First, the task is concurrent identification: the atrial-fibrillation record follows the sinus-rhythm record within 31 days (median 5 days), which is not long-horizon prediction of future onset. Second, a mean AUROC of about 0.60 to 0.64 is a feasibility-level result on a small public cohort. It is appropriate to a methods and benchmarking contribution and not to a screening tool. Third, the study rests on a single public database, and external validation on an independent database is the natural next step. Fourth, the ten control draws share their positives and are therefore not independent replicates, which is why the pooled bootstrap resamples patients jointly across draws.

**Future work.** Incident-AF prediction over longer horizons needs dated-outcome cohorts with follow-up beyond the days covered here, and assembling such a cohort is the most direct extension. On the method side, interpretable extensions of the block-term estimator, such as regularized or discriminant variants, could sharpen the signature while keeping the maps readable.

# 5 Conclusion

We applied an intrinsically interpretable tensor model to the identification of atrial fibrillation in sinus rhythm on public data with age- and sex-matched controls. BTTR on the R-peak-aligned median beat reached a ten-draw mean AUROC of 0.597 on the AFIB cohort and 0.643 on the AFIB-or-flutter cohort, and no baseline had a higher mean AUROC. Over the ten draws its AUROC was higher than that of the CNN and the fixed-setting HuBERT-ECG probe, with pooled paired 95% CIs excluding zero in both cohorts. The comparisons with the tuned HuBERT-ECG probe and clinical features do not separate the methods, and its discrimination was comparable to that of the CP surrogate on the same tensor. Its decision is exposed as eight time-by-lead weight maps whose per-lead profile reproduces across control draws at Pearson $r = 0.857$, with weight distributed across the beat and the largest share in the T window. R-peak alignment is required for the model, with the pooled paired CI of the alignment effect excluding zero in both cohorts. We report the weight distribution as an observation and claim no mechanism.

# Declarations

**Funding.** This work was supported by the Flanders AI Research Program (FAIR), an Impulse Program of the Flemish Government (Flemish Ministry of Economy, Science and Innovation).
**Competing interests.** The authors declare no competing interests.
**Ethics approval.** This study is a secondary analysis of PTB-XL, a publicly available, fully de-identified electrocardiogram database, and did not involve new human-subjects research; no additional ethics approval was required.
**Consent to participate.** Not applicable, because the study analyzes a public de-identified database.
**Consent for publication.** Not applicable, because the study analyzes a public de-identified database.
**Data availability.** The PTB-XL database (version 1.0.3) is publicly available [20].
**Code availability.** The code to reproduce the cohorts, models, and analyses is available at `https://github.com/TheAxeC/bttr-af-ecg`. The repository also holds the record-level cohort and split manifests of all ten control draws for both cohorts (ecg_id, patient identifier, label, demographics and outer fold of every record), and one command regenerates all results from PTB-XL.

**Use of AI-assisted tools.** AI-based writing-assistance tools were used to help draft and edit the manuscript text. All scientific content, analyses, numerical results, and interpretations were produced and verified by the authors, who take full responsibility for the work.

# References


[1] M. A. Al-Masud, J. M. Lopez Alcaraz, and N. Strodthoff. Benchmarking ECG FMs: A reality check across clinical tasks. *arXiv preprint arXiv:2509.25095*, 2025. https://doi.org/10.48550/arXiv.2509.25095.

[2] Zachi I. Attia, Peter A. Noseworthy, Francisco Lopez-Jimenez, Samuel J. Asirvatham, Abhishek J. Deshmukh, Bernard J. Gersh, Rickey E. Carter, Xiaoxi Yao, Alejandro A. Rabinstein, Brad J. Erickson, Suraj Kapa, and Paul A. Friedman. An artificial intelligence-enabled ECG algorithm for the identification of patients with atrial fibrillation during sinus rhythm: a retrospective analysis of outcome prediction. *The Lancet*, 394(10201):861–867, 2019. https://doi.org/10.1016/S0140-6736(19)31721-0.

[3] Yong-Soo Baek, Sang-Chul Lee, Wonik Choi, and Dae-Hyeok Kim. A new deep learning algorithm of 12-lead electrocardiogram for identifying atrial fibrillation during sinus rhythm. *Scientific Reports*, 11(1):12818, 2021. https://doi.org/10.1038/s41598-021-92172-5.

[4] Edoardo Coppola, Mattia Savardi, Mauro Massussi, Marianna Adamo, Marco Metra, and Alberto Signoroni. HuBERT-ECG as a self-supervised foundation model for broad and scalable cardiac applications. *medRxiv*, 2024. https://doi.org/10.1101/2024.11.14.24317328. Preprint. Pretrained on 9.1M 12-lead ECGs; model: Edoardo-BS/hubert-ecg-small (HuggingFace).

[5] José Henrique de Morais Goulart, Pedro Marinho Ramos de Oliveira, Rodrigo Cabral Farias, Vicente Zarzoso, and Pierre Comon. Alternating group lasso for block-term tensor decomposition and application to ECG source separation. *IEEE Transactions on Signal Processing*, 68: 2682–2696, 2020. https://doi.org/10.1109/TSP.2020.2985591.

[6] Pedro Marinho R. de Oliveira and Vicente Zarzoso. Block term decomposition analysis in long segments of atrial fibrillation ECGs. In *XXXVI Simpósio Brasileiro de Telecomunicações e Processamento de Sinais (SBrT 2018)*, 2018. https://doi.org/10.14209/sbrt.2018.182.

[7] Axel Faes, Flavio Camarrone, and Marc M. Van Hulle. Single finger trajectory prediction from intracranial brain activity using block-term tensor regression with fast and automatic component extraction. *IEEE Transactions on Neural Networks and Learning Systems*, 35(7): 8897–8908, 2024. https://doi.org/10.1109/TNNLS.2022.3216589.

[8] Tove Hygrell, Fredrik Viberg, Erik Dahlberg, Peter H. Charlton, Katrin Kemp Gudmundsdottir, Jonathan Mant, Josef Lindman Hörnlund, and Emma Svennberg. An artificial intelligence-based model for prediction of atrial fibrillation from single-lead sinus rhythm electrocardiograms facilitating screening. *EP Europace*, 25(4):1332–1338, 2023. https://doi.org/10.1093/europace/euad036.

[9] Jun Lei, Yuxi Zhou, Xue Tian, Qinghao Zhao, Qi Zhang, Shijia Geng, Qingbo Wu, and Shenda Hong. A deep learning method for beat-level risk analysis and interpretation of atrial fibrillation patients during sinus rhythm. *Biomedical Signal Processing and Control*, 100:107028, 2025. https://doi.org/10.1016/j.bspc.2024.107028.

[10] Jared W. Magnani, Lei Zhu, Faye Lopez, Michael J. Pencina, Sunil K. Agarwal, Elsayed Z. Soliman, Emelia J. Benjamin, and Alvaro Alonso. P-wave indices and atrial fibrillation: cross-cohort assessments from the Framingham Heart Study (FHS) and Atherosclerosis Risk in Communities (ARIC) study. *American Heart Journal*, 169(1):53–61.e1, 2015. https://doi.org/10.1016/j.ahj.2014.10.009.

[11] Pietro Melzi, Ruben Tolosana, Alberto Cecconi, Ancor Sanz-Garcia, Guillermo J. Ortega, Luis Jesus Jimenez-Borreguero, and Ruben Vera-Rodriguez. Analyzing artificial intelligence systems for the prediction of atrial fibrillation from sinus-rhythm ECGs including demographics and feature visualization. *Scientific Reports*, 11(1):22786, 2021. https://doi.org/10.1038/s41598-021-02179-1.

[12] Aruna Mohan, Danne Elbers, Or Zilbershot, Fatemeh Afghah, and David Vorchheimer. Deciphering heartbeat signatures: a vision transformer approach to explainable atrial fibrillation detection from ECG signals. In *2024 46th Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*, pages 1–6, 2024. https://doi.org/10.1109/EMBC53108.2024.10782666.

[13] Sajad Mousavi, Fatemeh Afghah, and U. Rajendra Acharya. HAN-ECG: An interpretable atrial fibrillation detection model using hierarchical attention networks. *Computers in Biology and Medicine*, 127:104057, 2020. https://doi.org/10.1016/j.compbiomed.2020.104057.

[14] Panteleimon Pantelidis, Samuel Ruipérez-Campillo, Julia E. Vogt, Alexios Antonopoulos, Ioannis Gialamas, George E. Zakynthinos, Michael Spartalis, Polychronis Dilaveris, Jose Millet, Theodore G. Papaioannou, Evangelos Oikonomou, and Gerasimos Siasos. ECG-XPLAIM: eXPlainable locally-adaptive artificial intelligence model for arrhythmia detection from large-scale electrocardiogram data. *Frontiers in Cardiovascular Medicine*, 12:1659971, 2025. https://doi.org/10.3389/fcvm.2025.1659971.

[15] Lucas Plagwitz, Lucas Bickmann, Michael Fujarski, Alexander Brenner, Warnes Gobalakrishnan, Lars Eckardt, Antonius Büscher, and Julian Varghese. The Rlign algorithm for enhanced electrocardiogram analysis through heart rate-corrected ECG alignment for explainable classification and clustering. *European Heart Journal - Digital Health*, 7(5):ztag067, 2026. https://doi.org/10.1093/ehjdh/ztag067.

[16] Lucas N. Ribeiro, André L. F. de Almeida, and Vicente Zarzoso. Enhanced block term decomposition for atrial activity extraction in atrial fibrillation ECG. In *2016 IEEE Sensor Array and Multichannel Signal Processing Workshop (SAM)*, pages 1–5, 2016. https://doi.org/10.1109/SAM.2016.7569709.

[17] Amirsajjad Taleban, Rodney Sparapani, Patrick Noffke, Sharone Zlochiver, Qiang Lu, Michael E. Widlansky, and Jake Luo. Explainable artificial intelligence in electrocardiography: a systematic review. *Biomedical Signal Processing and Control*, 114:109325, 2026. https://doi.org/10.1016/j.bspc.2025.109325.

[18] Constantine Tarabanis, Vidya Koesmahargyo, Dimitrios Tachmatzidis, Vasileios Sousonis, Constantinos Bakogiannis, Robert Ronan, Scott A. Bernstein, Chirag Barbhaiya, David S. Park, Douglas S. Holmes, Alexander Kushnir, Felix Yang, Anthony Aizer, Larry A. Chinitz, Stylianos Tzeis, Vassilios Vassilikos, and Lior Jankelson. Artificial intelligence-enabled sinus electrocardiograms for the detection of paroxysmal atrial fibrillation benchmarked against

the CHARGE-AF score. *European Heart Journal - Digital Health*, 6(6):1134–1144, 2025. https://doi.org/10.1093/ehjdh/ztaf100.

[19] Arne Van Den Kerchove, Hakim Si-Mohammed, François Cabestaing, and Marc M. Van Hulle. BTTDA: block-term tensor discriminant analysis for brain–computer interfacing. *arXiv preprint arXiv:2511.04292*, 2025. https://doi.org/10.48550/arXiv.2511.04292.

[20] Patrick Wagner, Nils Strodthoff, Ralf-Dieter Bousseljot, Dieter Kreiseler, Fatima I. Lunze, Wojciech Samek, and Tobias Schaeffter. PTB-XL, a large publicly available electrocardiography dataset. *Scientific Data*, 7(1):154, 2020. https://doi.org/10.1038/s41597-020-0495-6.

[21] Neal Yuan, Grant Duffy, Sanket S. Dhruva, Adam Oesterle, Cara N. Pellegrini, John Theurer, Marzieh Vali, Paul A. Heidenreich, Salomeh Keyhani, and David Ouyang. Deep learning of electrocardiograms in sinus rhythm from US veterans to predict atrial fibrillation. *JAMA Cardiology*, 8(12):1131–1139, 2023. https://doi.org/10.1001/jamacardio.2023.3701.

# Author biographies

Dries Cornelissen holds an MSc in Computer Science from Hasselt University, with a master's thesis on BTTR for arrhythmia detection and prediction on sinus rhythms, supervised by Axel Faes.
Axel Faes (PhD, computational neuroscience, KU Leuven) is a postdoctoral researcher at the University of Twente working on foundation models and multi-modal federated learning for precision medicine.